

WHAT ARE GENES?

A gene is generally regarded as a unit of DNA needed to construct one protein. It consists of all the sections of DNA that code for all the amino acids for that protein. Usually, one gene is located on one chromosome. However, it may have several sections on the DNA molecule, each containing the code for one portion of the protein. Typically, lengths of DNA called introns and exons (see below) are both transcribed to form immature mRNA. The parts of mRNA made from the introns are then stripped out by the cell's molecular machinery, leaving mature mRNA for translation. There are also regulatory DNA sequences that code for their own proteins, affecting the rate of gene transcription.

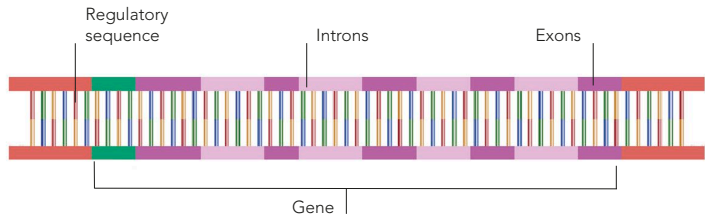


EYE COLOUR

Iris colour is affected by at least 15 genes, including OCA2 and HERC2, both sited on chromosome 15.

PARTS OF A GENE

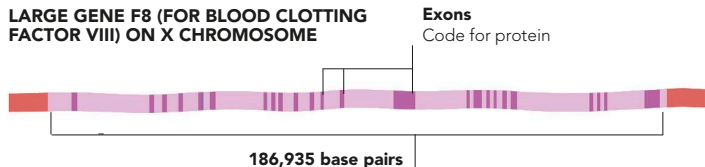
Regions called introns and exons both transcribe to form mRNAs for different portions of a protein. The lengths made from introns are then spliced out chemically, to leave exon-only portions, which go on to make the protein.



RANGE OF GENE SIZE

Genes vary enormously in their size, which is usually measured in numbers of base pairs. Small genes may be just a few hundred base pairs long, while others are measured in millions of base pairs. The gene for beta-globin is one of the smallest. It codes for part of the haemoglobin molecule. It is compared, right, with a larger gene.

LARGE GENE F8 (FOR BLOOD CLOTTING FACTOR VIII) ON X CHROMOSOME



SMALL GENE (FOR BETA-GLOBIN) ON CHROMOSOME 11

